

Functional Requirements

1 Client Functional Requirements

1.1 View Shops

The system shall display a list of all available airport shops.

Each shop shall display:

- shop name
- shop type

The client shall be able to select exactly one shop at a time.

After selecting a shop, the system shall display all items available in that shop.

Verification:

Select a shop and confirm that only that shop's items are displayed.

1.2 Sort Shops - with CustomerService team

The client shall be able to sort displayed shops by:

- name (ascending)
- rating (ascending)

Only one sorting criterion may be active at a time.

Verification:

Apply sorting options and confirm order.

1.3 Search Items

The system shall provide a search bar after a shop is selected.

The client shall be able to search for items by name.

The system shall display all items whose name contains the entered text (case-insensitive).

If no items match, the system shall display a “No items found” message.

Verification:

Search using partial and full item names.

1.4 Sort Items

The client shall be able to sort displayed items by:

- name (ascending)
- price (ascending)

Only one sorting criterion may be active at a time.

Verification:

Apply sorting options and confirm order.

1.5 Manage Cart

The client shall be able to:

- add items to the cart
- remove items from the cart
- change item quantity

The system shall prevent adding quantities greater than available stock.

The cart shall store:

- item name
- unit price
- quantity
- total price per item

Verification:

Attempt to add more items than available stock.

1.6 View Cart

The client shall be able to view cart contents at any time.

The cart view shall display:

- list of items
- quantity per item
- individual item total
- total cart price

Verification:

Add items and confirm totals are correct.

1.7 Reserve Cart

The client shall be able to reserve all items currently in the cart.

Reservation shall only succeed if sufficient stock exists.

When a reservation is confirmed:

- stock quantities shall decrease
- the reservation shall be stored in the system

The client shall be able to cancel a reservation.

Upon cancellation:

- stock quantities shall be restored
- the cart shall be cleared.

Verification:

Reserve items and confirm that stock quantities decrease accordingly.

Cancel the reservation and confirm that stock quantities are restored and the cart is cleared.

2 Administrator Functional Requirements

2.1 Manage Shops

The administrator shall be able to:

- add a shop
- update a shop
- delete a shop

Each shop shall include:

- name (unique)
- type

Deleting a shop shall remove its inventory.

2.2 Manage Shop Inventory

The administrator shall select a shop before managing inventory.

The administrator shall be able to:

- add an item to a shop
- remove an item from a shop
- update item price
- update stock quantity

The system shall ensure:

- stock ≥ 0
- price > 0

4 Non-Functional Requirements

The system shall prevent invalid input such as negative prices or stock values.

The system shall respond to user actions within 2 seconds under normal load.

The system shall maintain data consistency after reservations and cancellations.

The system shall support concurrent reservation without inconsistent stock updates.

The system shall maintain data persistence for shops, items, and reservations.